

Review

Prospects for summer cover crops in southern Australian semi-arid cropping systems

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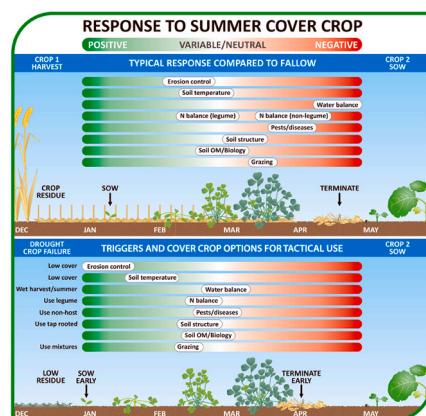
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HIGHLIGHTS

- Evidence for benefits of cover crops in Mediterranean-type climates is lacking.
- Opportunities to sow cover crops over the hot dry summer period are limited.
- Cover crop water use may reduce yields of subsequent cash crops.
- Tactical use of summer cover crops in targeted situations may be warranted.

GRAPHICAL ABSTRACT



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ABSTRACT

CONTEXT: Cover crops are widely promoted in temperate regions of the northern hemisphere to replace or reduce fallow periods between cash crops. The reported benefits (improved ground cover, reduced nitrate leaching, increased soil organic matter among others) have prompted widespread interest in cover cropping in other agro-climatic zones, including Mediterranean environments such as southern Australia, where the wet, mild winters and hot dry summers are a stark contrast to temperate environments.

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Soil health
Soil organic matter

OBJECTIVE: This review considers the reported benefits of cover crops in temperate environments in the context of cropping systems in the water-limited, Mediterranean climate in the grain-growing regions of southern Australia.

METHODS: Published studies on cover cropping were critically assessed to evaluate whether the reported impacts may be relevant to Mediterranean-type climates.

RESULTS AND CONCLUSIONS: Nitrate leaching is not generally problematic in southern Australian cropping systems due to low summer rainfall and high temperatures, and complete retention of crop stubbles can adequately protect the soil from erosion. Further, summer rainfall is low and sporadic which not only limits cover crop planting opportunities but means that water conserved over summer fallows contributes significantly to winter cash crop yield. Consequently, cover crop duration would need to be short (1–2 months) to balance the benefits of increased cover for water infiltration with transpiration loss during cover crop growth. Restricted cover crop biomass production due to water constraints and short fallow length will limit the potential value of the cover crop to increase soil organic matter, microbial activity, and biologically-fixed nitrogen contributions – all reported benefits for higher biomass cover crops in temperate environments. Appropriate tactical use of summer cover crops may arise in situations where (i) soil cover is reduced by winter crop failure or prolonged drought creating significant erosion risk, (ii) recalcitrant herbicide resistant weeds in winter crops require a chemical spring fallow and summer cover/forage crop, which could be grazed, (iii) full soil profiles at harvest and wet seasonal outlook provide opportunities for cover crop growth without significant pre-emptive competition, and (iv) sandy soil profiles with low water holding capacity combined with reliable winter rainfall minimise contributions of summer rainfall to winter crop yields. The value of more diverse cover crop species to soil health in cereal-brassica or cereal-brassica-legume cash crop rotations is not known.

SIGNIFICANCE: The rationale for widespread summer cover crop use in southern Australia is currently not compelling without more evidence for the scale and mechanisms of the anticipated benefits.

1. Introduction

Mediterranean climates are characterised by mild, wet winters and hot dry summers, and regions with these climates tend to be semi-arid and vulnerable to droughts (Seager et al., 2019). These climatic features are a key driver of the agricultural systems present in Australia's broad-acre cropping region south of 32°S latitude. Farms in this region typically receive an average 300–700 mm of rainfall annually (Fig. 1),

with high year-to-year variability. This variability is predicted to increase under a warming climate (Pendergrass et al., 2017). Owing to a lack of summer rainfall, the farming systems are dominated by dryland winter crop production, with a large proportion of 'mixed' farms with livestock that graze annual or perennial pastures phased in rotation with the crop sequence (Kirkegaard and van Rees, 2019).

The cropping systems of southern Australia are also characterised by a high adoption of minimum- or no-till practices (Llewellyn et al., 2012).

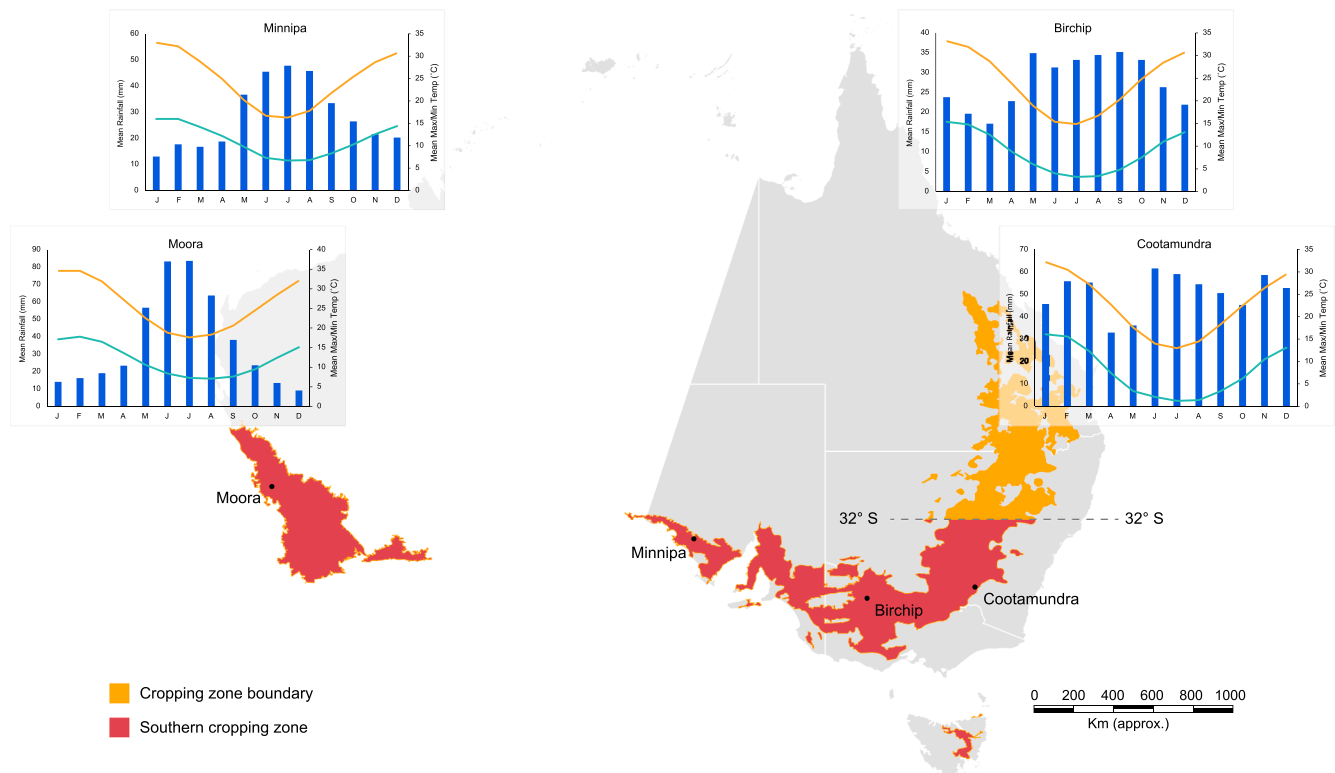


Fig. 1. The Australian broadacre cropping zone. The southern cropping zone is depicted in red and comprises the area below 32°S latitude in the east of the country and cropping regions of South Australia and Western Australia. The location and long-term average climatic conditions of four geographically spread towns within the southern cropping zone are shown to illustrate differences in rainfall patterns and total precipitation across the area. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Stubble is retained over the summer fallow period to protect the soil from erosion, and less than 5% of farmers reported burning stubbles in autumn (May) to avoid trash impediments at sowing (Umbers, 2017). The cropping sequences are dominated by cereals, owing to their yield stability under variable rainfall (Harries et al., 2020). However, break crops including canola (*Brassica napus* L.) and winter pulses are a key component of the cropping sequence in these no-till systems to minimise disease build up and allow rotation of herbicide groups to slow the rate of herbicide resistant weeds (Angus et al., 2015; Kirkegaard and van Rees, 2019; Kirkegaard et al., 2008).

An important trend in recent years for southern Australian cropping systems has been the emphasis on capturing summer rainfall to retain soil moisture to establish and grow the winter crops. Modelling studies across 37 locations in southern Australia suggest summer rainfall stored in the soil during the fallow period contributes an average 1 t ha^{-1} to wheat (*Triticum aestivum* L.) yields (Hunt and Kirkegaard, 2011). However, the contribution varies from as little as 0.1 t ha^{-1} in the northern zone of the Western Australian wheatbelt that is characterised by reliable in-season (winter) rainfall and sandy soils with low water holding capacity (WHC), to 2 t ha^{-1} in the central west of New South Wales where equi-seasonal rainfall and soils with high WHC increase the value of stored summer rainfall. Further improvements in water use efficiency (WUE) have been achieved with recent use of early-sown cereals and brassicas (March–April) with a vernalisation requirement (Hunt et al., 2021). While the degree to which crops can be sown earlier than their traditional autumn sowing window is subject to appropriate varieties and available moisture, warmer soil temperatures mean that plants establish quicker and have a longer vegetative period generating deeper root systems able to extract water from a greater depth (Flohr et al., 2020; Hunt et al., 2019). Some of these ‘dual-purpose’ crops can also be grazed during winter. There is little impact on wheat yields if animals are removed prior to the onset of the reproductive phase, so dual-

purpose systems provide flexibility, profitability and risk mitigation (Dove and Kirkegaard, 2014).

There is also growing interest in carbon (C) sequestration, ‘regenerative agriculture’, soil health and biodiversity within Australian broad-acre agricultural systems. This is partially due to the intensification of cropping that has occurred since the 1990s, which has reduced the area of pastures that have traditionally been a regenerative phase for soil fertility (see Pratley and Kirkegaard, 2019 and references therein). Growers in continuous cropping systems are seeking innovative options to maintain the soil, environmental and economic benefits in their farming systems. Interest in cover crops among some grain growers has been stimulated by the reported benefits of winter cover crops in temperate environments in the northern hemisphere (Abdalla et al., 2019; Kaye and Quemada, 2017). Cover crops are single or mixed-species plantings that are not harvested for yield and typically replace or reduce a traditional fallow phase between cash crops (Snapp et al., 2005). In the context of southern Australian farming systems, cover crops could be planted as short-term winter or summer crops in low rainfall areas where long fallowing is practiced, or during the summer fallow period between consecutive winter cash crops, which is typical of the majority of cropping systems in southern Australia. Green or brown manure crops that replace cash crops, or short or long-term pasture phases, are not considered to be cover crops in this context because they are not grown in a typical fallow period. In addition, the use of dual-purpose crops for grazing and grain has resulted in additional options for diversity within the existing farming system. The planting of mixed-species (such as vetch [*Vicia villosa* Roth], tillage radish [*Raphanus sativus* L.]) with an early sown, vernalisation-dependent wheat, results in increased quality and quantity of biomass for grazing animals. The additional species are then removed with a selective herbicide prior to cessation of grazing and taking the wheat through to grain. This practice is a form of (temporary) intercropping as opposed to cover cropping. In

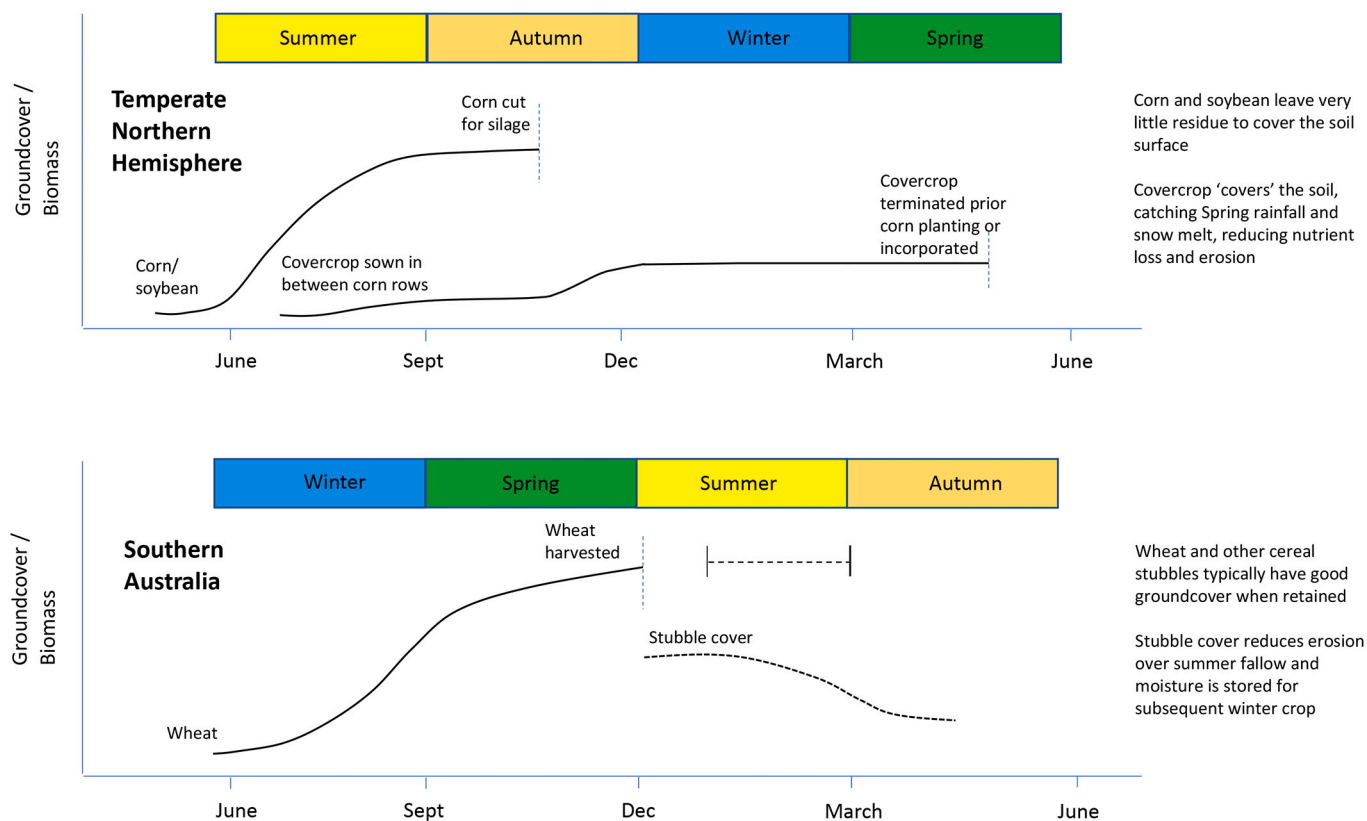


Fig. 2. Representation of cover cropping in temperate cropping systems in the northern hemisphere compared with southern Australian cropping systems. The period of the proposed summer cover crop growth is shown by the horizontal dotted line.

this paper, we focus on summer cover crops grown between successive winter cash crops (Fig. 2). These short-term summer crops may be used for grazing in mixed farming systems, and for the purpose of this paper we still consider them to be cover crops, provided they are sown in the summer fallow between two consecutive winter cash crops.

Decades of research on winter cover crops in temperate regions of the northern hemisphere have identified numerous potential benefits including increases in soil organic matter (Blanco-Canqui et al., 2015; Kaye and Quemada, 2017; Lal, 2015; Poeplau and Don, 2015; Ruis and Blanco-Canqui, 2017), additional nitrogen (N) inputs (Thorup-Kristensen et al., 2003), improved weed control (Osipitan et al., 2018), and reductions in soil erosion (Blanco-Canqui et al., 2015; Dabney, 1998; Dabney et al., 2001), nitrate (NO₃) leaching (Abdalla et al., 2019; Thapa et al., 2018; Thorup-Kristensen et al., 2003) and nitrous oxide (N₂O) emissions (Basche et al., 2014; Kaye and Quemada, 2017), with few apparent trade-offs. The decision to plant cover crops, the choice of cover crop species and the duration of the cover crop phase are usually determined based on one or two selected objectives of the land manager, although secondary benefits may also arise. The main benefits of cover crops in temperate growing regions in the northern hemisphere are summarised in Table 1. Overall, these specific benefits can result in wider improvements in soil and water quality with limited impact, or positive impacts, on subsequent cash crop yields (Abdalla et al., 2019; Blanco-Canqui et al., 2015; Marcillo and Miguez, 2017; Snapp et al., 2005; Tonitto et al., 2006).

In contrast to the large number of studies from temperate environments of Europe and North America where the water use by cover crops has limited impact on the growth of subsequent cash crops, fewer studies have examined the potential for cover crops in water-limited environments. A review by Unger and Vigil (1998) and subsequent studies by Nielsen and Vigil (2005) and Nielsen et al. (2016) provided several examples of cash crop yield reduction following summer cover crops compared to traditional fallows. The studies highlight the need for timely establishment and removal of cover crops to avoid excessive water use in water-limited environments. The cropping systems described for the semi-arid Great Plains of USA involving traditional long fallows with moves towards two crops per three years (wheat-sorghum [*Sorghum bicolor* (L.) Moench] -fallow) in southern semi-arid areas of the Great Plains (Unger and Vigil, 1998) share similarities to Australia's semi-arid, subtropical cropping zone. Research in Australia's northern grain region prompted by concerns over erosion due to a lack of groundcover during long fallows and in wide crop row systems have demonstrated the importance of timely sowing and termination of cover crops to generate net benefits to the system (Whish et al., 2009). Though sharing similar principles, neither of these cropping systems are directly applicable to the dryland cropping systems in southern Australia where annual winter crops are separated by a short 4–6-month fallow period during hot, dry summers.

There has been limited scientific experimentation on summer cover crops in southern Australia to inform the growing interest among farmers aroused by the reported benefits promoted by the international scientific and farming communities. However, given the clear differences between southern Australian cropping systems and the temperate cropping systems in the northern hemisphere where most benefits of cover cropping are reported (Fig. 2), it is worth exploring the rationale for the proposed benefits of cover crops in the context of cropping systems in Mediterranean environments with high adoption of minimum tillage practices. In this review, we examine the likely benefits and trade-offs of summer cover crops in the dryland cropping environment of southern Australia where hot dry summers, mild winters and unreliable or ineffective rainfall contrast significantly with the environments where most cover crop studies reported in the international literature have been conducted.

Table 1

Utility of summer cover crops in southern Australian farming systems. Symbols X, XX, XXX and XXXX represent no or little benefit, some potential benefit, consistent benefits, and strong benefits, respectively.

Benefit	Evidence or strong rationale for proposed benefits ^a		Justification of rationale for southern Australian systems
	Northern Hemisphere Temperate climate	Southern Australian Mediterranean-type climate	
Increased groundcover and reduced erosion	XXXX 1,2	XX	Cover crops may play a role where winter crops (e.g. legumes) have low residue levels or when winter crops fail in drought years.
Increased soil organic matter	XXXX 2,3,4,5	X	Unlikely to benefit Australian systems due to water limitations. Any biomass produced by cover crops may be offset by lower biomass in winter crops.
Reduction in NO ₃ leaching	XXXX 1,6	XX	Nitrate leaching over the summer period is rare due to hot temperatures and low rainfall. However, deep drainage following wet autumns on sandy soils may be avoided by drying profile over summer with cover crops.
Nitrogen fixation and N supply to subsequent crops	XXX 5,7,8	X	Low summer rainfall will likely limit potential biomass production, and therefore N fixation.
Improved soil water conditions and soil temperatures	XXX 1	X	Crop residues aid capture of summer rainfall in southern Australia, and dry soils, as opposed to wet soils, usually limit sowing of winter crops.
Improved soil biological function	XXXX 9,10	XX	Low biomass production by summer cover crops likely limits impacts on soil biology, but the use of mixed species cover crops may increase microbial diversity.
Amelioration of compaction and subsoil constraints	XXX 11	X	Existing crop rotations generally sufficient to minimise disease issues associated with monocultures.
	X	XX	Summer cover crops are unlikely to penetrate deep into subsoils in dry soil conditions; winter species with taproots (e.g. canola) or perennial species grown during the pasture phase provide a greater opportunity.

(continued on next page)

Table 1 (continued)

Benefit	Evidence or strong rationale for proposed benefits*		Justification of rationale for southern Australian systems
	Northern Hemisphere Temperate climate	Southern Australian Mediterranean-type climate	
Amelioration of non-wetting sands			Summer cover crops may create water entry pathways in non-wetting sands. Likely insufficient moisture and growing period to sufficient cover crop biomass to suppress summer weeds. Impact on winter crop pests is unknown. Reduced herbicide use over summer may lower residual herbicide loads in soil, but this benefit may be outweighed by excessive moisture use by cover crops. While summer cover crops may provide high value feed, there is no evidence this option is preferable to the existing ley pasture phases and grazing of summer stubbles used in mixed farming systems.
Control of weeds and pests	XX 12	X	
Reduced herbicide use	X	XX	
Fodder for livestock	X	XX	

¹Dabney et al., 2001; ²Blanco-Canqui et al., 2015 ³Lal, 2015; ⁴Poeplau and Don, 2015; ⁵Kaye and Quemada, 2017; ⁶Thapa et al., 2018; ⁷Frye et al., 1988; ⁸Dabney et al., 2011; ⁹Jian et al., 2020; ¹⁰Kim et al., 2020; ¹¹Blanco-Canqui and Ruis, 2020; ¹²Osipitan et al., 2018.

* Where possible the references provided as evidence for a benefit of cover crops in temperate farming systems in the northern hemisphere are comprehensive reviews or meta-analyses as opposed to a single study that reported that benefit in a particular location or season.

2. Potential to establish summer cover crops

Before exploring the rationale and potential role of summer cover crops in southern Australian farming systems, it is worthwhile examining the potential opportunity to sow summer cover crops in Mediterranean-type climates, since a lack of rain in the typically dry, hot summer period may limit opportunities to successfully establish the summer cover crop. Using the definition of a 'sowing opportunity' as a period where the sum of rainfall received in a 7-day period is greater than the evaporation for that period (Flohr et al., 2021; Unkovich, 2010), we modelled the probabilities of cover crop sowing opportunities from December 1 – February 28 (summer) across eight locations in southern Australia using historical weather data from 1971 to 2020 (Table 2) obtained from the SILO data base (Jeffrey et al., 2001). Cootamundra, NSW, which has a mean annual rainfall of 638 mm with equi-seasonal distribution (see Fig. 1) had the highest percentage of years where a sowing opportunity occurred in the 3-month summer window (65%; Table 2). The median for the first sowing opportunity at Cootamundra occurred on December 15, which would allow for cover crop establishment early in the fallow period and consequently time for the cover crop to accumulate biomass. In contrast, in the winter-dominant rainfall areas of Western Australia, a sowing opportunity arose in only 16% of years (Moora – see Fig. 1) or 22% of years (Ardath), with the median sowing opportunity arising in late January (Table 2). Opportunities for early establishment and thus biomass accumulation of summer cover crops in these environments are rare. Thus, even where tactical use of summer cover crops may be beneficial for improving ground cover, there is no guarantee that conditions will be suitable for timely establishment.

3. Potential role of summer cover crops in cropping systems of southern Australia

Most of the reported benefits of cover crops in the northern hemisphere, including reduced NO₃⁻ leaching or improved C or N inputs, are closely related to the level of biomass production, which is in turn linked to water use. In water-limited environments, water conserved in the soil during the summer fallow phase can make a substantial contribution to the grain yield of subsequent cash crops (Hunt and Kirkegaard, 2011). Summer cover crops, like summer weeds, are likely to significantly reduce the soil water and N available to subsequent crops (Hunt et al., 2013), as has been seen when cover crops have been introduced into winter wheat-fallow systems in the semi-arid central Great Plains of the

Table 2

Summer cover crop sowing opportunities from December 1 – February 28 across eight locations in southern Australia using historical weather data (1971–2020) obtained from the SILO data base (Jeffrey et al., 2001). 25th–75th percentiles represent the frequency distribution of where sowing opportunities fall between the December–February window at each location. A sowing opportunity was defined as a period where the sum of rainfall received in a 7-day period is greater than the evaporation for that period (Unkovich, 2010, Flohr et al., 2021).

Location	State	Met Station Number	Mean annual rainfall (mm)	% years with sowing opportunity	Percentile when sowing opportunity does occur			Total number of years with a sowing opportunity
					25th	median	75th	
Birchip	Vic	77,088	345	43	15-Dec	22-Jan	8-Feb	21
Yarrawonga	Vic	81,057	503	51	10-Dec	1-Jan	28-Jan	26
Condobolin	NSW	50,059	453	47	17-Dec	5-Jan	30-Jan	28
Cootamundra	NSW	73,009	638	65	6-Dec	15-Dec	21-Jan	38
Minnipa	SA	18,052	321	10	8-Dec	17-Dec	19-Feb	5
Bordertown	SA	25,501	461	31	2-Dec	3-Jan	3-Feb	15
Moora	WA	8091	415	16	17-Jan	30-Jan	19-Feb	8
Ardath	WA	10,503	340	22	20-Jan	30-Jan	18-Feb	11

United States (Nielsen and Vigil, 2005; Nielsen et al., 2016). The overall effect of summer cover crops on subsequent cash crop yield in water-limited environments will depend on the net effects of water, N balance and other system parameters that may be influenced by cover crops (see below). Longer-term impacts beyond the immediate cash crop (weed seed banks, disease inoculum, soil structure, soil health) may also arise. We are unaware of any peer-reviewed field data on the mechanisms or net impact of summer cover crops on subsequent cash crop yields in Mediterranean-type environments. As a consequence, we focus the following discussion on the potential cover crop impacts on soil and agronomic parameters that are likely to interact to affect subsequent cash crop yields in Mediterranean-type cropping environments of southern Australia.

4. Increased ground cover and reduced erosion

Compared with bare fallows, cover crops can reduce erosion by increasing water infiltration and reducing velocity of water runoff due to physical cover crop presence (Dabney, 1998; Dabney et al., 2001), improving soil aggregate strength due to C inputs (Lal, 2015), and protecting the soil from rain drop impact or wind erosion (Blanco-Canqui et al., 2013). As a result, there is consistent evidence from studies

in temperate environments that winter cover crops contribute to a reduction in soil erosion (see review by Blanco-Canqui et al., 2015).

In water-limited environments, retention of cash crop residue can achieve the same erosion benefits as cover crops without depletion of soil water reserves (Unger and Vigil, 1998). However, where long fallows are practiced to conserve soil moisture, low ground cover and increased erosion potential can occur following crops with low residual stubble (e.g. grain legumes such as lentils [*Lens culinaris* Medik]), following crops grown on wide row spacing or following tillage for weed control (Nielsen and Vigil, 2005; Whish et al., 2009). In these circumstances, cover crops may play an important role in soil erosion control. For example, Blanco-Canqui et al. (2015) reported significant reductions in soil erosion following a spring pea (*Pisum sativum* L.) cover crop compared to fallow in a semi-arid (462 mm rainfall) area of Kansas where long fallows are typically practised. Given the water limitations in these environments, the ideal cover crops are often those that provide rapid ground cover without excessive depletion of soil moisture. In the long fallow cropping systems of Australia's semi-arid, subtropical cropping zone, the spring-sown C4 grass Japanese millet (*Echinochloa esculenta* L.) outperformed legume cover crops because it accumulated biomass faster, achieved ground cover faster, and had more persistent residue than the legumes that were tested (Wunsch et al., 2017). The

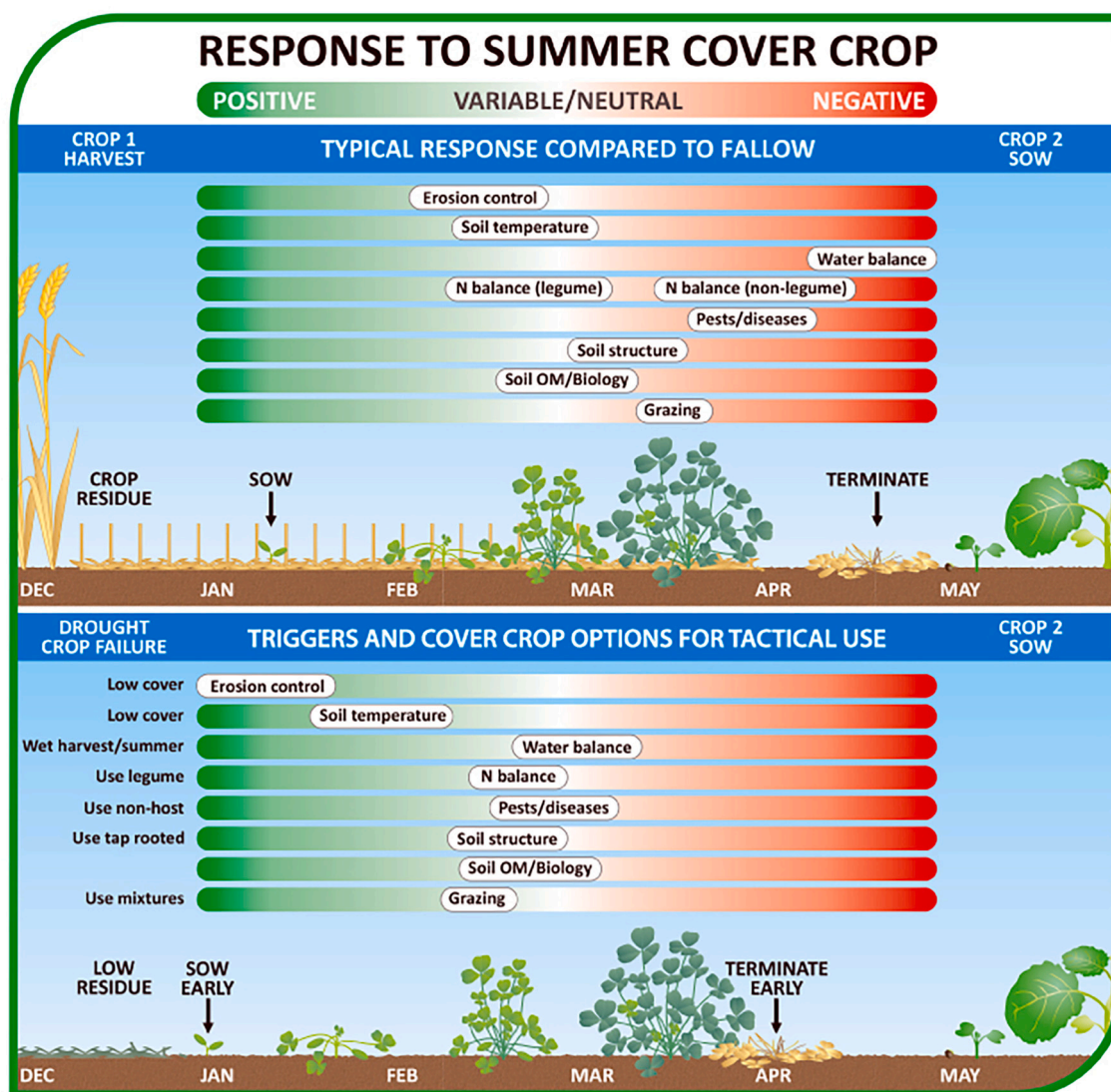


Fig. 3. Proposed productivity and environmental outcomes from summer cover crops in typical seasons (top panel) and when used tactically (lower panel) in southern Australian cropping systems.

rapid degradation of legume residues also has negative effects on rain infiltration into soils after termination of cover crops because higher residue levels typically enhance infiltration (Bell et al., 2021).

While summer rainfall is typically lower in frequency and intensity in southern Australia than in Australia's semi-arid, subtropical cropping zone, data suggest late summer rainfall in Wagga Wagga still has the potential to cause high rates of erosion (Scott et al., 2010). However, standing stubbles remain during this period to mitigate the risk of erosion (Fig. 2). The benefits of cover crops for erosion control where adequate residue loads are present are therefore questionable, particularly with the risk of excessive use of stored fallow water (Fig. 3). During drier seasons as few as 2–3% of farmers subsequently burn stubbles late in autumn prior to sowing due to lighter stubble loads, while in wetter years with heavy stubble loads 35–40% of fields may be burnt in autumn prior to sowing (Scott et al., 2013). When low residue cover occurs after dry cropping seasons or following low-residue crops including winter pulses (Bell et al., 2021), summer cover crops may reduce the risk of erosion. The predicted increase in rainfall extremes under climate change is likely to increase the chance of soil erosion in southern Australia (Liddicoat et al., 2012). The impact of summer cover crops on soil erosion risk will be highly context specific, depending on existing stubble loads, soil type and topography and influenced by the effect of sowing the summer cover crop on existing stubble configuration and degradation rates. Nonetheless, cover crops may be beneficial in protecting soil from erosion in summer and early autumn in circumstances where cover levels are low e.g. following droughts, failed crops or low residue winter pulse or oilseed crops (Fig. 3).

5. Improved water management

In temperate environments where planting of summer cash crops can be impeded by wet field conditions (limited field trafficability), water use by winter cover crops can dry the soil profile sufficiently to enable timely sowing of summer cash crops (Dabney et al., 2001). While trafficability can occasionally be an issue in water-limited environments, by far the most dominant problem is a lack of soil moisture at sowing for the cash crop following hot, dry summers (Table 2), so the timing of sowing and termination of cover crops is critical to the water available for subsequent crops (Unger and Vigil, 1998). In water-limited environments where long fallowing is practiced, the results of cover cropping on soil-water relations (and subsequent crop yields) have been variable, ranging from benefits (Erbacher et al., 2019) to no effect (O'Dea et al., 2013; Whish et al., 2009; Zentner et al., 2004) to significant detrimental effects (Nielsen and Vigil, 2005; Vigil and Nielsen, 1998). For example, water use by legume cover crops in a wheat-fallow system in the central Great Plains was detrimental to subsequent wheat yields regardless of the time of cover crop removal (Nielsen and Vigil, 2005; Vigil and Nielsen, 1998). In contrast, spring-sown French millet (*Panicum miliaceum* L.) cover crops did not significantly reduce soil profile water at subsequent wheat sowing in a long fallow system in Australia's semi-arid subtropics provided cover crops were terminated 30–70 days after sowing (Whish et al., 2009). Validated crop simulation studies suggested early-sown French millet cover crops would only have adverse effects on subsequent wheat yields in this environment in 2% of years if cover crops were removed after 50% ground cover had been attained (Whish et al., 2009). The fact that French millet cover crops did not significantly deplete soil water compared to fallow controls in the study of Whish et al. (2009) was attributed to greater capture of summer rainfall in the cover crop treatments to offset water use by the cover crop. The ability to capture more rainfall depends on a range of factors that often interact to influence infiltration, which include the quantity of existing ground cover, soil type, tillage, topography, and the amount and intensity of rainfall (Scott et al., 2013; Verburg et al., 2012). Inconsistencies in results from long fallow systems may be in part driven by differences in these parameters. The results of Whish et al. (2009) also relate to northern Australian grain-growing environments with significant

summer rainfall and on clay soils with high storage potential and high erosion risk.

Despite the lack of specific summer cover crop studies on soil water dynamics in southern Australia, there has been a significant body of work since 2010 investigating the impacts of summer fallow management on winter cash crops, including the impacts of summer weed control, stubble cover and grazing management. Since growing a summer cover crop effectively mimics uncontrolled summer fallow weeds, the outcomes of that work are highly relevant here. Hunt and Kirkegaard (2011) used simulation to predict the impact of uncontrolled summer weeds (including wheat volunteers) on the water dynamics and yield of subsequent wheat at 37 sites across the southern Australian wheatbelt. Conserving summer fallow rain by controlling summer fallow weeds resulted in average wheat yield increases of 33% (range 3 to 72%) and approximately one third of water limited potential yield in southern Australia was attributable to stored summer fallow rain (Hunt and Kirkegaard, 2011). The value of controlling summer weeds was highest (72%) in areas such as the red loam soils in central western NSW which combine significant (50%) summer rainfall, soils able to store summer rainfall and unreliable rainfall in spring. The lowest value was on sandy soils in northwest Western Australia with a lower proportion of summer rainfall, soils with low water holding capacity and in areas of reliable winter rainfall. In a series of subsequent field experiments, these predictions were verified with average yield gains of 32% with strict summer weed control (Kirkegaard et al., 2014). In a study where the timing of the weed control was explored (mimicking the duration of a proposed cover crop) delaying weed control by 3 weeks significantly reduced subsequent yields by around 16% compared with 39% with no summer weed control. Further field experiments also revealed that the yield gains from summer fallow weed control weren't entirely due to water conservation, but also differences in measured mineral N ($\text{NO}_3^- + \text{NH}_4^+$) content of the soil (Hunt et al., 2013). In general, where subsequent crop seasons were dry, the water savings dominated the response while in wet seasons the additional N saving drove the yield response. As water and N always co-limit yield in semi-arid environments (Sadras et al., 2012) it is likely both factors will operate across most seasons.

Cover crops can enhance water infiltration by preventing surface soil sealing, particularly in tilled soil, and by decreasing the rate of surface runoff which enhances water infiltration (reviewed by Dabney, 1998) - an effect that can also be achieved by retention of crop stubbles (Fig. 2). In the same way summer weed control mimics cover crop impacts, so too is the work on stubble retention instructive. The large existing literature and more recent work on the benefits of retaining crop stubble over summer on soil water conservation and subsequent crop yields in southern Australia (see review by Flower et al., 2019) are consistent and compelling. The major effect of (stubble) cover is to reduce run-off and increase water infiltration by reducing structural damage by raindrop impact which causes surface sealing (Scott et al., 2013). The beneficial effects on infiltration reach a maximum at around 70% soil cover and higher levels of cover (or residue) have minimal further impact on this process (Scott et al., 2013). Higher rates of cover can slow, but not stop, evaporative loss from the soil surface, so that during southern Australian summers with very high evaporative rates, covered and uncovered soils may both dry to an equal extent in prolonged dry periods (Scott et al., 2013; Verburg et al., 2012). As evaporative losses decline in autumn the covered soil will retain higher moisture content for a much longer period of time, which many farmers hope to exploit by leaving high stubble loads in order to sow crops earlier (Hunt et al., 2021). The effect of cover in any specific circumstance is determined by soil type and the distribution and intensity of the rainfall as described by the "pulse paradigm" reported by Verburg et al. (2012). Kirkegaard et al. (2014) report that of 29 field experiments that compared retained vs. removed crop residue during the summer fallow period in southern Australia, 17 showed no significant effect of residue retention on subsequent crop yield, three showed a positive effect (mean 9%) and seven a negative effect (mean -12%). Residue cover had a strong effect on increasing soil water

infiltration (Hunt et al., 2016), but the effect on evaporation was to slow, but not stop evaporative loss in summer so that after long dry periods the surface soils would still dry (Verburg et al., 2012). Yield benefits only ensued where summer fallow rain was intense, residues improved infiltration beyond the evaporative zone and thus soil water storage was increased. More detailed experiments determined that a threshold of residues was required to ensure rainfall infiltration, but beyond this there was little benefit (Hunt et al., 2016).

Concerns had been raised that grazing of stubble over summer would degrade soil structure and reduce water infiltration and yield of subsequent crops. However, both experimental (Hunt et al., 2016) and simulation evidence (Bell et al., 2011; Stavi et al., 2015, 2016) demonstrated little if any effect of grazing provided stubble thresholds achieving 70% cover were maintained.

Taken together, these recent studies on the management of weeds, residue and grazing in summer fallows in southern Australia suggest that on average, the trade-offs of growing summer cover crops in terms of water and N supply to following crops will be large, while the benefits of the increased cover above that required for maximum infiltration (70%) will be small and sporadic (Table 1; Fig. 3). As an opportunistic practice, a summer cover crop would presumably be best applied where the soil water profile is full, and surface cover is low at harvest (e.g. after a legume and/or wet harvest period). In this situation the surface water used by a cover crop would likely be lost by evaporation anyway, and the cover crop residue after termination may slow evaporative loss in late summer and early autumn to facilitate early sowing (Fig. 3). However, this assumes the roots of the cover crop do not actively extract water from depths below the evaporative zone, using stored soil water otherwise available for subsequent cash crops. The period between cover crop termination and sowing the next cash crop would also require management to maximise the net benefits to the system, given the higher yield potential of earlier-sown cash crops (Hunt et al., 2021). The aim would be to facilitate the early sowing opportunity by specifically targeting the preservation of surface water in early autumn. In this case it is the timing and location (i.e. within the seedbed) of the additional soil water, rather than the absolute amount of additional water, that may be of significant benefit. Management strategies that conserve seed bed moisture are increasingly important in response to fewer sowing opportunities for timely crop establishment due to autumn rainfall decline in south-eastern Australia (Flohr et al., 2021).

6. Increased soil organic matter

Where the inclusion of a cover crop has no effect or improves the yields of subsequent cash crops (e.g. Marciello and Miguez, 2017), cover cropping can increase annual C inputs into soils, which may result in higher soil C levels over time. For example, the use of winter rye (*Secale cereale* L.) as a cover crop between summer corn (*Zea mays* L.) and soybean (*Glycine max* L. Merr.) crops in the mid-west USA led to increases of soil organic matter (SOM) and particulate organic matter (POM) of 15% and 44% in the 0–5 cm zone over an 11-year period compared to a fallow (Moore et al., 2014). As such, there is a strong consensus in review papers that in temperate environments cover cropping (mainly with winter rye as a cover crop) has significant potential to increase topsoil C stocks compared to fallow over time (Blanco-Canqui et al., 2015; Kaye and Quemada, 2017; Lal, 2015; McClelland et al., 2021; Poeplau and Don, 2015; Ruis and Blanco-Canqui, 2017).. However, cover crops do not always increase SOM in semi-arid environments, even when supplementary irrigation is used (Acharya et al., 2022; Thapa et al., 2022). Further, increases in SOM are also contingent upon adequate soil fertility to sustain microbial requirements for N, phosphorus (P) and sulfur (S) (Kirkby et al., 2011, 2016). While likely of less importance in areas of Europe and North America with young and fertile soils, the generally poor background fertility of the highly weathered soils of the southern Australian cropping zone mean that SOM accrual is often limited by nutrient availability. As demonstrated

by Kirkby et al. (2016), increasing SOM requires adequate nutrients along with the C inputs into the system. The type of cover crop or mixture used will influence the potential for soil C sequestration because of differences in the quantity and quality of the residues in relation to the rates of microbial degradation and humification efficiency according to the stoichiometry of nutrients and other compounds within the tissues (Coonan et al., 2020).

Cover crops may confer additional benefits for topsoil C in temperate environments by reducing soil loss from water and wind erosion (see above). It is worth noting, however, that recent evidence from a long-term (>20 years) experiment in California suggests that topsoil C increases under cover crop treatments may be offset by decreases in C in lower (30–200 cm) layers (Tautges et al., 2019). Whether the findings of Tautges et al. (2019) have broader implications beyond the particular conditions of their experimental site is yet to be seen.

In water-limited environments, the inclusion of cover crops in a rotation can lead to reductions in the yield of subsequent cash crops in some seasons (Mubvumba et al., 2021; Nielsen et al., 2016; Rosa et al., 2021; Unger and Vigil, 1998). The net impact of cover crops on soil C inputs in water-limited environments is therefore unclear since C inputs into the system from cover crops may be offset by a reduction in C inputs into the soil in the cash crop phase. In the water-limited Mediterranean-type environment of southern Australia it is also possible that biomass contributions of summer cover crops may be offset by biomass reductions of winter cash crops, through the uneven (and unpredictable) partitioning of total annual rainfall across the two growth cycles, but published field data are currently lacking. We note that a recent meta-analysis of the potential for cover crops to build soil C (Abdalla et al., 2019) used data from the Mediterranean-type environment of Wagga Wagga, Australia (Zhou et al., 2011) to represent 7 of the total 43 data points on SOC. However, the methods section of the original study by Zhou et al. (2011) indicates that the ‘cover crops’ were grown over the winter cash crop period, not the traditional fallow period, and soil C was measured following termination of the cover crops compared to a fallow control. Given that the cover crops were effectively green manure crops replacing the winter cash crops, and that soil C measurements were taken soon after the termination of cover crops and compared to a fallow control, the data do not accurately reflect the potential for cover crops to improve long-term soil C in Mediterranean-type environments. While field studies are clearly needed to investigate annual C dynamics with summer cover crop inclusion into winter cash crop rotations in southern Australia, soil water limitations would suggest increasing C inputs *per se* into the soil would not be the primary aim of cover cropping in these environments (Table 1; Fig. 3).

7. Reduced NO₃⁻ leaching

One of the key functions of winter cover crops in temperate cropping systems is to reduce NO₃⁻ leaching (Kaye and Quemada, 2017; Kaye et al., 2019; Thorup-Kristensen et al., 2003), with a recent meta-analysis suggesting non-leguminous winter cover crops reduce NO₃⁻ leaching by around 56% compared to fallow controls (Thapa et al., 2018). Review papers also indicate that legume/non-legume cover crop mixes can be equally as proficient at reducing NO₃⁻ leaching as pure non-legume cover crop stands (Dabney et al., 2001; Kaye et al., 2019; Thapa et al., 2018).

Despite the reduction in NO₃⁻ leaching by winter cover crops in temperate environments, their utility to reduce NO₃⁻ leaching in most Mediterranean-type environments is questionable for two reasons. Firstly, greater reduction in NO₃⁻ leaching is driven by high cover crop biomass production (Finney et al., 2016), which will be linked to higher water use in Mediterranean-type climates. Secondly, in southern Australia where winters are cold and wet, and summers are hot and dry, the few studies that have measured NO₃⁻ leaching indicate that leaching or lateral subsurface movement of N generally occurs from late autumn to early spring (Anderson et al., 1998; Ridley et al., 2001; Smith et al., 1998). The lack of NO₃⁻ leaching observed over summer is attributed to

the negative water balance where evapotranspiration is typically much greater than precipitation (Smith et al., 1998). It is possible, however, that water stored in the subsoil over summer may drain deeper beyond the rooting zone of winter crops following wet autumn or early winter conditions, particularly on deep sands (Dolling et al., 2006). However, it is this same stored subsoil moisture that can make a significant contribution to winter cash crop yields in drier seasons (Hunt and Kirkegaard, 2011). As pointed out by Angus and Grace (2016), large-scale summer flooding events occur sporadically across semi-arid Australia resulting in complete inundation of soils for many weeks. In these instances, it is highly likely that nearly all NO_3^- would be denitrified (Angus and Grace, 2016) and the accumulation of N in cover crop tissues may reduce denitrification losses. On the whole, however, reducing NO_3^- leaching losses would not be a primary aim of summer cover cropping in southern Australia (Table 1).

8. Nitrogen supply to subsequent crops and lower N_2O emissions

In temperate cropping systems, winter legume cover crops can contribute significant quantities of N into the system, although much of this N does not become immediately available to subsequent crops due to low rates of mineralisation in dryland environments (Cabello Leiva, 2020). Similarly, non-legume cover crops such as radish can reduce soil NO_3^- leaching by accumulating N in biomass over the traditional fallow period, but this N is not mineralised rapidly enough to become available to the subsequent cash crop (Ruark et al., 2018). More recent studies indicate that mixed cover crops with a legume component can also contribute substantial amounts of N to the system while simultaneously reducing NO_3^- leaching (e.g. De Notaris et al., 2021; Kaye et al., 2019). Indeed, the inclusion of a legume in a cover crop mix can be essential to maximise cover crop biomass because in some situations growth of non-legumes can become limited by a lack of N (Dabney et al., 2001). The N contributions from winter legume cover crops in temperate regions may lower cropping system N_2O emissions by offsetting N fertiliser requirements. While a meta-analysis found no clear evidence that cover crops consistently reduce N_2O emissions from the soil *per se* (Basche et al., 2014), a recent review indicates that the main impact of winter cover crops on the greenhouse gas footprint of cropping systems is the reduction in the use of fertiliser N, a result of N contributions from legume components of cover crop mixes (Kaye and Quemada, 2017). While the N fertiliser requirements of the cash crop grown immediately after a cover crop may not decline due to slow mineralisation of N in cover crop residues (Cabello Leiva, 2020; Ruark et al., 2018), it is logical that N fertiliser requirements for a cropping system would decline over the long term if cover crops can reduce NO_3^- leaching and/or fix N_2 (legumes).

Field experiments in Australia's semi-arid, subtropical cropping zone, where long fallowing is practiced to conserve soil moisture, found that while legume monoculture cover crops could fix up to 25 kg N ha^{-1} in shoots during a 90-day period in spring prior to a winter wheat crop, they were slower to accumulate biomass (groundcover) and used more water than Japanese millet and were predicted to reduce wheat yields in >60% of years simulated by APSIM modelling (Wunsch et al., 2017). While it therefore seems unlikely that legume cover crops grown in water-limited environments can produce the large (> 100 kg N ha^{-1}) inputs reported in temperate environments (Tonitto et al., 2006), it is worth noting that the average fertiliser N input into Australian wheat and oilseed crops is only 45 kg N ha^{-1} (Angus and Grace, 2016). Assuming a root/shoot N ratio of 60–90-d-old legumes of roughly 1:2 (Rose et al., 2018a), the legume cover crops in the study of Wunsch et al. (2017) could have provided around 38 kg N ha^{-1} in total. While this equates to around 85% of the average amount of fertiliser N applied to Australian winter cereal or oilseed crops, there is no guarantee that the N in legume residues would mineralise rapidly enough to be available to the subsequent winter crop. In the Mediterranean-type climate of

southern Australia the fallow period is shorter (typically December–April; Fig. 2) and it is unlikely that cover crops could be grown beyond 90 days without affecting subsequent winter cash crop yields. The extent to which the legume components of a summer cover crop could make a significant contribution to the N nutrition of the subsequent winter crop remains unknown in such environments, but given the short length of the fallow and that appreciable N_2 fixation by legumes typically occurs beyond 40 days after sowing (Rose et al., 2018a), it seems unlikely that N contributions will be the dominant factor in a farmer's decision to use a cover crop in these environments (Table 1; Fig. 3).

9. Enhanced soil biological function (soil resilience, disease and pest suppression)

Research from temperate environments suggest that cover cropping is beneficial for soil aggregation, water infiltration and soil microbial communities over the long term (Jian et al., 2020; Kim et al., 2020), but these benefits are predominantly linked to the overall annual increases in C inputs into the soil with cover crops in these environments (Lal, 2015). If cover cropping in water-limited or Mediterranean-type environments does not increase net soil C inputs but simply spreads the timing of inputs more evenly across the year, the associated beneficial functions mediated by soil biology may not persist or accumulate. Liu et al. (2005) found that non-leguminous winter cover crops grown for 8 months could increase aggregate stability significantly in a degraded silt loam even without significant increases in total soil organic C. It is therefore possible that small changes caused by cover cropping could affect other beneficial soil biological functions without measurable changes in organic C.

Kim et al. (2020) conducted a meta-analysis on cover crop impacts on soil microbial abundance, activity and diversity. Sufficient information was only available for relatively general measures including microbial biomass C and N, enzyme activities and diversity, as opposed to more specific functions such as N-fixation, nutrient mobilisation, or disease suppression. They found strong evidence that cover crops increased microbial biomass C and N by 25% compared to fallow control plots. Respiration, β -glucosidase activity and phosphatase activity were also higher on average in cover crop treatments, by 35, 18 and 14%, respectively. The magnitude of the responses to cover cropping was also significantly moderated by climate, soil type, termination method (mechanical or chemical), tillage and the timing of the sampling. Interestingly, increases in microbial abundance and activity in response to cover cropping were smaller in no-till systems, which may have already established more abundant and active soil microbiomes, and therefore the effect of cover crops was less pronounced in these systems (Kim et al., 2020). Overall, the effects on microbial abundance and activity were more pronounced than diversity, suggesting the primary influence of cover crops was through greater organic matter input, rather than alterations to niches (e.g. new habitats, diversified food sources) that drive microbial diversity. Unfortunately, the seasonal phase of the cover crop (e.g. summer or winter cover crop) was not one of the factors reported or analysed by Kim et al. (2020), so it is difficult to know whether these general trends would be as strong, or even consistent, in the winter-dominant cash crop systems of semi-arid Australia with a history of stubble retention. Further, given the time of sampling had a strong influence on results, a key question in Australian systems is whether any changes to soil biology that may occur in late summer following a summer cover crop would persist long enough to have any effect during the subsequent cash crop phase or beyond.

Furthermore, as Kim et al. (2020) point out, more research is needed into the effect of cover crops on specific microbial processes to help make inferences about how and why cover crops enhance crop productivity, and in which circumstances the benefits outweigh the trade-offs. To this end, a number of studies have examined the effect of

cover crops on specific functions, including the effect on arbuscular mycorrhizal fungi (AMF) and associated influence on crop nutrition and soil structure, N₂ fixation and disease suppression. Bowles et al. (2017) conducted a meta-analysis on the effects of winter/fall cover crops on AMF colonisation of the subsequent summer cash crop, primarily corn or soybean. They found that relative to winter fallows, cover crops increased colonisation of summer cash crop roots by 28.5% (95% CI:12.1–47.4). Colonisation was significantly greater when the cover crop was an AMF host and was greater if the cover crop was a legume rather than a graminoid or non-legume dicot (Bowles et al., 2017).

Although fewer in number, some studies examining summer cover crop/winter cash crop systems support the findings of Bowles et al. (2017) regarding improved AMF colonisation generated by cover crops. Galvez et al. (1995) reported increased AMF in corn after a wheat crop in NE USA when a hairy vetch winter cover crop was grown as opposed to a normal fallow, though notably plots were only 1 m². Spore counts of vesicular mycorrhizal fungi (VAM; range of species) were significantly higher where a hairy vetch cover crop was grown. Somenahally et al. (2018) also showed that winter wheat cash crops had higher levels of AMF colonisation (by 30–70%) after a summer cowpea (*Vigna unguiculata*) cover crop than a summer fallow on the southern Great Plains of the USA. Interestingly, the cover crop treatments did not increase SOC, implying that the effect on mycorrhizal colonisation was independent of the greater C inputs *per se*. This reinforces the notion that additional benefits related to AMF, such as improved soil structure (Rillig and Mummey, 2006), may also arise in the absence of measurable increases in soil OC. Together, these studies provide strong evidence that the presence of cover crops in otherwise fallow fields can increase AMF colonisation in the roots of subsequent cash crops. In northern Australia after long fallows (12 months or greater), the lack of AMF colonisation on Vertosols in particular can cause poor crop nutrition and serious yield declines, termed ‘long-fallow’ disorder (Thompson, 1987). In these cases, introduction of a cover crop for a part of the fallow may improve the nutrition of subsequent cash crops through greater AMF colonisation. Indeed, Erbacher et al. (2019) reported that various cover crops grown in a long-fallow situation, as opposed to the plant-free fallow, actually increased plant-available water and grain yield of a subsequent wheat crop in a drier-than-average season. Wheat yields were greater than what could be attributed solely to the increase in plant available water (PAW), but other factors that could have contributed to the positive response were not explored. Considering that the late-terminated cover crops out-performed the early-terminated crops, it is plausible that maintenance of active mycorrhizae via living roots close to sowing of the subsequent crop led to greater AMF colonisation of wheat seedling roots and plant growth promotion. Whether or not similar mechanisms would significantly contribute to winter cash crop yield grown under more temperate climates and different soil types in southern Australia is not known, but it is also important to note that AMF colonisation is considered less important in southern Australian cropping systems than in northern Australian grain systems (Ryan et al., 2002) and indeed may even be detrimental to wheat yields under low soil P conditions (Ryan and Graham, 2018; Ryan and Kirkegaard, 2012). Interestingly, two of the major broad-leaf rotation crops with wheat in southern Australia, canola (*B. napus*) and lupin (*Lupinus angustifolia*), are actually non-mycorrhizal, yet generate large yield benefits in following wheat crops (Kirkegaard et al., 2008; Angus et al., 2015; McBeath et al., 2015). Thus, cover cropping may not lead to improved winter crop yields despite increased AMF colonisation.

There is also a plethora of specific examples (often pot studies) showing the effect of prospective cover crop species on a particular pathogen or group of pathogens or nematodes. Importantly, for effective disease management, the selected cover crop should not be a host to the same pathogens as the cash crop (Larkin, 2015) (Fig. 3). This follows the same principles when using crop rotation (i.e. rotation of the main crop) for disease management. If disease management is a key aim, crops known to produce nematode and/or pathogen suppressive chemicals, i.

e. ‘biofumigant’ crops, appear to provide better outcomes (Couëdel et al., 2019; Larkin, 2015; Acharya et al., 2021). This includes members of the Brassicaceae family, which can be grown as winter cover crops, and sorghum/sudangrass species (*Sorghum* sp.), which are summer-dominant. For maximum benefit of biofumigant crops, incorporation as a green manure is recommended, as this maximises the release of the bioactive chemicals: glucosinolates/isothiocyanates in the case of Brassicaceae and cyanide in the case of sorghums (see Dutta et al. (2019); Matthiessen and Kirkegaard (2006)). A recent meta-analysis suggests that, of the Brassicaceae, *Raphanus* and *Eruca* spp. provide superior biofumigation to *Brassica* spp. (Morris et al., 2020). The benefits of an additional cover crop to a system where crop rotation, including the use of *Brassica* spp. (i.e. canola in southern Australia), is the norm are not particularly clear. Further, many cropping systems in southern Australia include a multi-year pasture phase, during which a range of annual and perennial species are grown and grazed by livestock. Detheridge et al. (2016) compared the legacy effect of a 3-year forage phase comprised of different, single-species swards (either ryegrass [*Lolium multiflorum*] ± N, chicory (*Cichorium intybus* L.) red clover [*Trifolium pratense* L.] or white clover [*Trifolium repens* L.]) on soil fungal communities under subsequent cereal crops (spring wheat followed by winter barley [*Hordeum vulgare* L.]). Although the authors identified the forage phase as a ‘cover crop’ in the title of their paper, it is more comparable to the pasture phase in mixed farming systems of southern Australia rather than what we have defined here as a summer cover crop. Nevertheless, Detheridge et al. (2016) showed that the choice of forage species had a significant effect on the soil fungal community, with ryegrass (regardless of N fertiliser) favouring higher levels of AMF fungal guilds compared with other forage species; whereas soil under red clover harboured higher levels of dark septate endophytes. Levels of pathogenic fungi were highest under white clover, coinciding with higher legacy soil NO₃ concentrations; but disease symptoms were not observed in cereals under any preceding forage treatment. Differences in soil fungal communities under the different forage species treatments persisted from the end of the forage phase through to the end of the second cereal crop, but diminished over time. Whether mixed-species summer cover crops during the 4–5-year cropping phase provide added benefit beyond the diversity from the pasture phase is not known.

10. Amelioration of soil compaction, subsoil constraints and non-wetting sands

Soils may be naturally strong, or may develop physical constraints to root growth following compaction by livestock or machinery or where tillage equipment leads to formation of hard pans in soils (Hamza and Anderson, 2005). While there is good evidence that crop roots can preferentially exploit ‘biopores’ created by previous crops (e.g. Rasse and Smucker, 1998; White and Kirkegaard, 2010), there is limited published evidence that particular plant species, or the use of cover crops *per se*, can create more biopores to facilitate root growth through compact soil layers or dense subsoils. Such assumptions for Australian southern systems were challenged by Cresswell and Kirkegaard (1995) who found little evidence that tap-rooted canola crops created useful biopores in compact sub-surface soils, while studies on the same duplex soils found perennial pasture species such as lucerne (*Medicago sativa*) and phalaris (*Phalaris aquatica*) could improve the number of subsoil biopores (McCallum et al., 2004).

Subsequent studies in Europe have shown that cover crops with large root diameter such as lucerne and chicory were capable of creating a greater number of biopores in the subsoil (Han et al., 2015). Such biopores can be used as preferential pathways for root growth of following cash crops such as wheat, barley and canola (Athmann et al., 2013; Huang et al., 2020; Kautz et al., 2013; Perkons et al., 2014). In temperate regions with lower bulk density in deep soil layers, some evidence suggests that the improved root growth via biopores increased accessibility to subsoil water (Gaiser et al., 2012) and nutrients (Han et al.,

2021) indicating that the roots were not confined inside the pores. In contrast, the clumping model developed in Australia (Passioura, 1991) suggests that restricted root growth inside pores may lead to poor soil-root contact where root growth is restricted to biopores and cracks in the highly compacted subsoils in Australia (White and Kirkegaard, 2010). However, direct measurement on the cover crop-mediated biopore genesis and its effects on cash crop root growth and resource acquisition under water-limited environment are rare and should be verified with further studies.

Winter cover crops grown in temperate cropping systems in the northern hemisphere provide a significant opportunity to generate root biomass in the subsoil, because the cover crops are grown over a period of many months in generally wet soil conditions with corresponding lower soil strength (Williams and Weil, 2004). In contrast, cover crops in southern Australian cropping systems are likely to be grown for a short duration (6–10 weeks) over the hot, dry summer months when soils are typically dry, with associated high soil strength, providing limited scope for the cover crops to penetrate compacted layers or dense subsoils regardless of the species grown (Cresswell and Kirkegaard, 1995). A pasture phase, where plants grow across multiple seasons and are therefore more likely to experience wetter soil conditions at some stage during the growth phase, provides a better opportunity for roots to penetrate deep into subsoils in these environments (Cresswell and Kirkegaard, 1995).

The potential for cover crops to reduce the impacts of non-wetting sands, which affect more than 10 million ha of land in southern Australia (Roper et al., 2015), may warrant further investigation. Sowing cash crops on or adjacent to previous crop rows has been shown to improve seedling emergence, likely due to improved water infiltration into soils through channels created by prior crops (Roper et al., 2015). Summer cover crops in the long fallow systems of Australia's northern cropping zone have been shown to improve water infiltration from intense summer rain events (Whish et al., 2009), and it is possible that residues from summer cover crops may also provide water entry pathways into soil in autumn to improve crop establishment on non-wetting sands in southern Australia. Further research is needed to examine whether summer cover crops can alleviate the conditions associated with non-wetting sands and whether the benefits would outweigh the issue of water use by the cover crop. The practicalities of establishing a summer cover crop in non-wetting sands also require investigation.

11. Impacts on weeds and pests

A recent meta-analysis of 46 field studies indicated that on the whole, cover crops can suppress weeds during the cover crop growth phase and during the early part of the cash crop growth phase (Osipitan et al., 2018). However, the vast majority of the 46 studies examined winter cover crops in environments that are not typically classified as water-limited. Weed suppression in these studies was therefore associated with the biomass produced by cover crops and subsequent sowing of cash crops soon after cover crop termination, where the bulky residues of the cover crops could suppress weeds during the early stages of the summer cash crop. Experiments in south-west Western Australia have examined the impact of winter cover crops (replacement of a winter cash crop) on weed dynamics (Flower et al., 2012), but we are unaware of any studies that have investigated the specific impact of summer cover crops on weed dynamics during the summer or subsequent winter cash crop phase. While the spectrum of weeds present over the summer fallow period is distinctly different to the winter (in-crop) weed spectrum in southern Australian cropping systems, summer weeds use stored moisture that could be used by subsequent cash crops (Hunt and Kirkegaard, 2011). Given the water limitations in southern Australia, and tendency in water-limited environments to terminate summer cover crops months prior to sowing subsequent crops to allow time for soil water to accumulate (Whish et al., 2009), it is unlikely that sufficient summer cover crop residue would be present in Autumn at

sowing in southern Australia to provide substantive suppression of weeds in the winter cash crop. However, a diverse range of herbicides are typically used to control weeds over the summer fallow period in southern Australia (Harries et al., 2020) and integration of summer cover crops may enable a reduction in the number or types of specific herbicide applications used. Given the patterns of herbicide use over summer tend to be specific to given regions, it is not possible to make generalisations about how herbicide usage over the fallow period may change with any integration of summer cover crops. Nonetheless, it is possible that there may be a reduction in herbicide use or change in usage pattern with any adoption of summer cover crops in southern Australia.

While relatively few studies have examined effects of cover crops on insect dynamics in broadacre crops (e.g. Bowers et al., 2020; Everts et al., 2006; Koch et al., 2015; Laub and Luna, 1992; Lundgren and Fergen, 2010), these studies generally focused on winter cover crop effects on summer cash crops. A feature of many winter cover cropping systems is termination of cover crops often occurs within weeks of sowing the summer cash crop (e.g. Bowers et al., 2020) or even after sowing of summer cash crops (e.g. Koch et al., 2015). A key strategy for the management of pests including aphids and mites that are vectors of important viral disease in cereals, oilseeds and pulse crops, and diseases including cereal rusts in southern Australian cropping systems, is the maintenance of a clean summer fallow to minimise the 'green bridge' (volunteer cereals and weeds that emerge over summer). Thus, it has been suggested that cover crops should be terminated at least two weeks before planting of cash crops to avoid pest carry over (Couëdel et al., 2019). Given that summer cover crops are likely to use excessive amounts of soil moisture should they be grown up to the time of winter crop planting in southern Australia, they are likely to be terminated well before winter crops are sown in late autumn (Fig. 3). In the absence of field data on summer cover crops in Mediterranean-type climates, little is currently known about what role cover crops, or specific cover crop species, may play in the suppression or carryover of specific pests or beneficial insects for winter cash crops.

12. Reduced herbicide use

The reliance on herbicides for weed control in no-till cropping systems is associated with several challenges, including the development of resistant weed populations (Beckie et al., 2019), potential carryover and/or accumulation of herbicide residues in soils (Van Zwieten et al., 2016), and potential off-site environmental impacts (Bish et al., 2021). While the use of knockdown and residual herbicides during the summer fallow may have limited consequences for the development of herbicide resistant winter weed populations, fallow spray applications contribute to the overall herbicide load in soils (Rose et al., 2022). Studies from Western Australia indicate that glyphosate is the most widely used herbicide, and it is predominantly used for fallow and pre-sowing weed control as opposed to in-crop (glyphosate tolerant crop) weed control (Harries et al., 2020). As a result of its widespread use, residues of both glyphosate and its breakdown product AMPA are widespread in both 0–10 cm and 10–20 cm layers of soil in Australian cropping soils (Rose et al., 2022). While there is limited evidence of detrimental effects of glyphosate use or soil residues on soil biota (Nguyen et al., 2016; Rose et al., 2016), the residues can be toxic to subsequent crops in sandy soils when phosphate fertilisers are applied at sowing (Rose et al., 2018b). Interestingly, modelling studies suggest summer fallow rain is of little benefit to winter crop yields in sandy soils with low water holding capacity in areas of reliable winter rainfall (Hunt and Kirkegaard, 2011). If a summer cover crop that is terminated by a single herbicide application replaces a clean fallow that receives multiple herbicide applications over the summer period (Table 1), benefits may arise through the lower herbicide load present in soil at sowing of winter cash crops and reduced time and cost required for summer spray operations. Crucially, however, the establishment of summer cover crops requires summer rainfall, and

this can be particularly unreliable in these environments (Table 2). A further consideration is that areas with sandy soils and reliable winter rainfall are potentially suited to 'pasture cropping' where winter cash crops are seeded into C4 perennial grasses that are grazed over summer and are generally dormant in winter (Lawes et al., 2014). If pasture cropping can be successfully undertaken in these areas, it may be a more profitable option than annual seeding of summer cover crops to achieve similar outcomes.

13. Grazed summer forages

The economics of integrating cover crops in southern Australian cropping systems may be more favourable in mixed farming enterprises where the cover crops could provide valuable summer feed for livestock. However, as with pasture cropping, the value of summer feed vs the trade-off with winter crop yields will depend on the type of livestock enterprise and balance of land allocated to cropping vs livestock production (Finlayson et al., 2012). The return on investment for cover crops as a feed source is also questionable given that the cost of establishment (seed and labour/diesel) for cover crops that provide one or two grazing events is likely similar to the cost of establishing a perennial (ley) pasture that provides feed for 3–4 years or a permanent C4 perennial pasture in areas where pasture cropping may be viable. Reliance on summer cover crops, as opposed to perennial pastures, to fill summer feed gaps is also risky in many southern Australian cropping regions because of limited sowing opportunities in these environments (Table 2). Thus, while summer cover crops may be used tactically to provide summer forage (Fig. 3), there is limited economic rationale for adopting summer cover crops over perennial pastures in most mixed farming operations in southern Australia (Table 1).

14. Conclusions

From this review of potential benefits of cover crops (summarised in Table 1; Fig. 3), there is no compelling evidence or rationale to suggest that the clear benefits of cover crops observed in temperate environment can provide regular substantive benefits in southern Australian cropping systems to justify the costs and potential trade-offs involved. In the absence of livestock, the costs of establishing summer cover crops would need to be offset by either environmental benefits (reduced soil erosion, reduced NO₃⁻ leaching, increased soil C) or increased system productivity by enhanced soil function, overcoming soil physical constraints, improving soil N status or minimising the build-up of herbicide residues in soil. Some of the potential benefits may arise from opportunistic cover cropping to restore soil cover after drought, failed crops or low residue crops in seasons where soil water is in good supply and seasonal precipitation forecasts are promising. Many of the proposed benefits of cover crops related to changes in soil physical or biological characteristics may take many years to emerge and quantify, or to translate to improved productivity and are likely linked to cover crop biomass and water use. Currently, no long-term studies that quantify soil changes under systems involving regular or opportunistic use of summer cover crops compared to a summer fallow are available on which to base solid conclusions. Pre-experimental modelling studies could explore the likely trade-offs between summer biomass production and winter cash crop yields to better identify the sites and seasonal conditions in which there is sufficient scope for summer cover crop use. A value proposition would be required to justify seed companies investing in making suitable varieties available for commercial use.

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Declaration of Competing Interest

The authors declare no conflict of interest.

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